HISTORY AND EVOLUTION OF AI

LESSON 2

A TIMELINE OF ARTIFICIAL INTELLIGENCE

The beginnings of AI: 1950s

Alan Turing

At a time when computing power was still largely reliant on human brains, the British mathematician Alan Turing imagined a machine capable of advancing far past its original programming. To Turing, a computing machine would initially be coded to work according to that program but could expand beyond its original functions.

At the time, Turing lacked the technology to prove his theory because computing machines had not advanced to that point, but he's credited with conceptualizing artificial intelligence before it came to be called that. He also developed a means for assessing whether a machine thinks on par with a human, which he called "the imitation game" but is now more popularly called "the Turing test."

AMERICAN ASSOCIATION OF ARTIFICIAL INTELLIGENCE FOUNDED
AFTER THE DARTMOUTH CONFERENCE IN THE 1950S, AI RESEARCH BEGAN SPRINGING UP AT VENERABLE INSTITUTIONS LIKE MIT, STANFORD, AND CARNEGIE MELLON. THE INSTRUMENTAL FIGURES BEHIND THAT WORK NEEDED OPPORTUNITIES TO SHARE INFORMATION, IDEAS, AND DISCOVERIES. TO THAT END, THE INTERNATIONAL JOINT CONFERENCE ON AI WAS HELD IN 1977 AND AGAIN IN 1979, BUT A MORE COHESIVE SOCIETY HAD YET TO ARISE

Early AI excitement quiets: 1980s-1990s

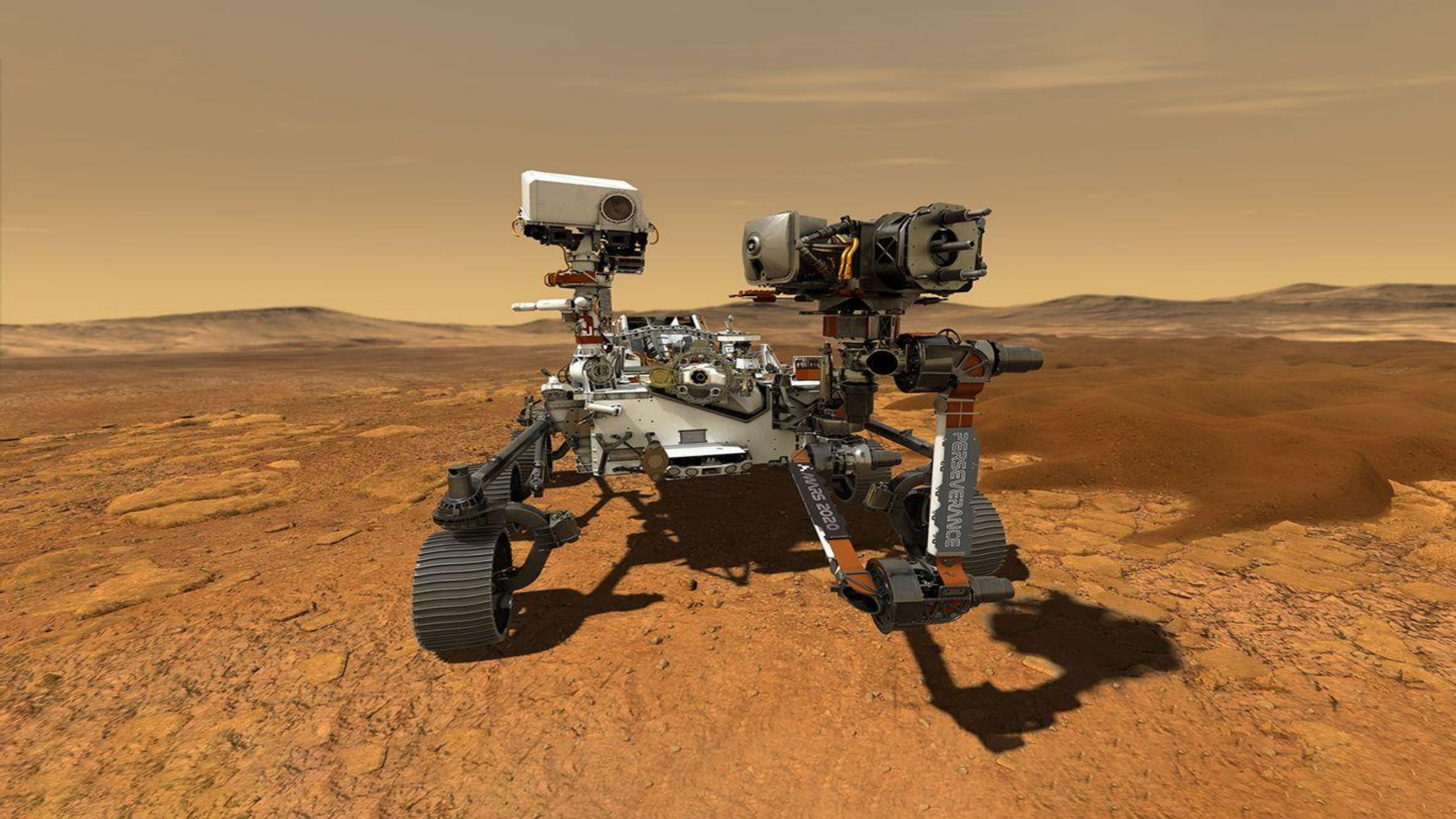
First driverless car
Ernst Dickmanns, a scientist working in Germany, invented the first self-driving car in 1986. Technically a Mercedes van that had been outfitted with a computer system and sensors to read the environment, the vehicle could only drive on roads without other cars and passengers.

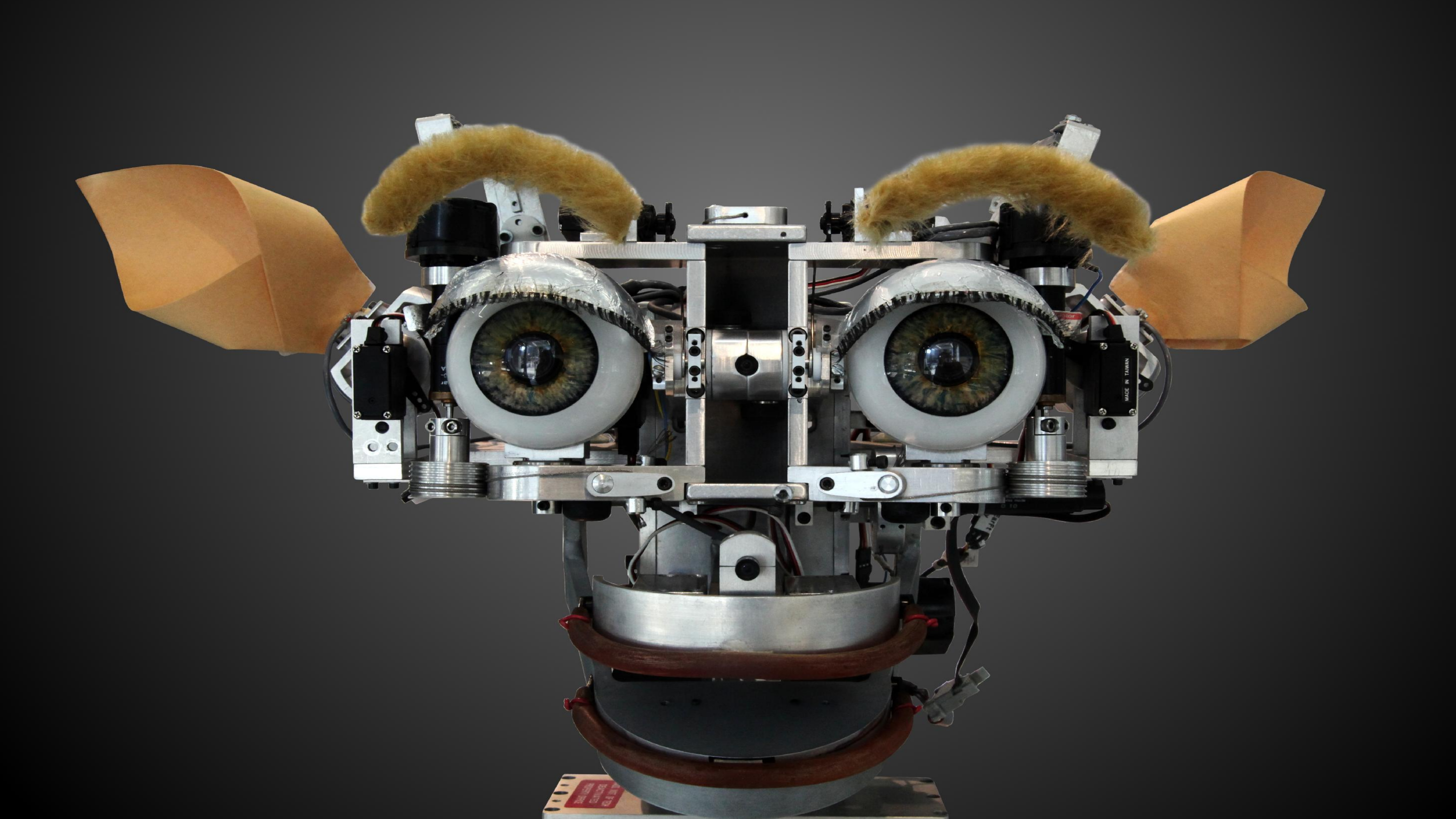
Deep Blue
In 1996, IBM had its computer system Deep Blue—a chess-playing computer program—compete against then-world chess champion Gary Kasparov in a six-game match-up. At the time, Deep Blue won only one of the six games, but the following year, it won the rematch. In fact, it took only 19 moves to win the final game.

AI growth: 2000-2019

Kismet
You can trace the research for Kismet, a “social robot” capable of identifying and simulating human emotions, back to 1997, but the project came to fruition in 2000. Created in MIT’s Artificial Intelligence Laboratory and helmed by Dr. Cynthia Breazeal, Kismet contained sensors, a microphone, and programming that outlined “human emotion processes.” All of this helped the robot read and mimic a range of feelings.

NASA rovers are robotic vehicles designed to explore the surfaces of other planets, moons, or celestial bodies. They are part of NASA's efforts to study the solar system and beyond, providing invaluable insights into the conditions, geology, and potential for life on other worlds.


PERSEVERANCE
MARS 2020


MADE IN TAIWAN

Siri and Alexa

During a presentation about its iPhone product in 2011, Apple showcased a new feature: a virtual assistant named Siri. Three years later, Amazon released its proprietary virtual assistant named Alexa. Both had natural language processing capabilities that could understand a spoken question and respond with an answer.

Yet, they still contained limitations. Known as "command-and-control systems," Siri and Alexa are programmed to understand a lengthy list of questions but cannot answer anything that falls outside their purview.

SOPHIA CITIZENSHIP

ROBOTICS MADE A MAJOR LEAP FORWARD FROM THE EARLY DAYS OF KISMET WHEN THE HONG KONG-BASED COMPANY HANSON ROBOTICS CREATED SOPHIA, A “HUMAN-LIKE ROBOT” CAPABLE OF FACIAL EXPRESSIONS, JOKES, AND CONVERSATION IN 2016. THANKS TO HER INNOVATIVE AI AND ABILITY TO INTERFACE WITH HUMANS, SOPHIA BECAME A WORLDWIDE PHENOMENON AND WOULD REGULARLY APPEAR ON TALK SHOWS, INCLUDING LATE-NIGHT PROGRAMS LIKE *THE TONIGHT SHOW*.

https://youtu.be/yaL5ZMvRRqE?si=T2kUAbF_GxEUaiBs

https://youtu.be/eSj80Zr6TEE?si=0whF1dgmsZNMWPJZ

https://youtu.be/RkeweRU_iLg?si=L35aNLxdb5t_d1I0

https://www.coursera.org/articles/benefits-of-ai?trk_ref=relatedArticlesCard

https://www.coursera.org/articles/types-of-ai?trk_ref=relatedArticlesCard

https://www.coursera.org/articles/generative-ai-applications?trk_ref=relatedArticlesCard

What is Chat-GPT?

Chat-GPT is like a really smart robot that can talk, just like a human. It's not a physical robot, but a program you can chat with using your computer or phone.

- Imagine if you had a super brain that read tons of books, websites, and stories. ChatGPT is trained in that way, so it knows a lot about many topics.
- You can ask it questions, get help with homework, write stories, or even play games with it!

What is Open-AI?

Open-AI is the company that made Chat-GPT. Think of it like a team of really smart scientists and programmers who work together to create cool tools and technologies that make computers smarter.

Their goal is to:

1. Create Artificial Intelligence (AI) that can help people.
2. Make AI safe and useful for everyone.

Must-watch videos

Humanoid robots

https://youtu.be/a34ZpoKP7hg?si=SsYYn3hey1uUWR4m

https://youtu.be/KvN3JXICzdM?si=opz0S6IQP3oPxYhN

https://youtu.be/gTSFBFmRJVs?si=1WZIStvgLCiCUTZD

https://youtu.be/uUMsZMWfnXE?si=Lu6it2P2cPEk2pfY